SQL PRACTICE WORKBOOK

100 SQL Practice Questions

Basic to Advanced — WHERE, GROUP BY, HAVING, Filtering, Set Operators, JOINs, Subqueries and Aggregate Functions

REFERENCE

Dataset Tables

Customers	CustomerID, FirstName, LastName, Country, Score
Employees	EmployeeID, FirstName, LastName, Department, BirthDate, Gender, Salary, ManagerID
Orders	OrderID, ProductID, CustomerID, SalesPersonID, OrderDate, ShipDate, OrderStatus, Quantity, Sales
OrdersArchive	Same structure as Orders
Products	ProductID, Product, Category, Price

PART A

WHERE, GROUP BY, HAVING & Filtering

Q1–Q10

Q1. From Customers, display customers whose Score is greater than 500. Show CustomerID, FirstName, Country and Score.

Q2. Find employees whose Salary is greater than 50,000 and whose Department is Marketing. Display their names, department and salary.

Q3. Find orders whose OrderStatus is Shipped or Delivered. Display OrderID, OrderStatus and Sales.

Q4. Display products whose Category is either Accessories or Clothing.

Q5. Find employees whose salary is between 55,000 and 80,000, inclusive.

Q6. Find customers whose FirstName starts with the letter M.

Q7. Find records where LastName is missing. Display CustomerID and FirstName.

Q8. Count the number of orders for each OrderStatus. Display the status and order count.

Q9. Calculate total sales and average sales for each ProductID.

Q10. Find CustomerID values whose total Sales is greater than 30. Display CustomerID and total sales.

PART B

SQL JOINs

Q11–Q30

Q11. INNER JOIN Orders and Customers. Display OrderID, CustomerID, customer FirstName and Sales for matching records.

Q12. INNER JOIN Orders and Products. Display OrderID, product name, category, quantity and sales.

Q13. Join Orders and Employees using SalesPersonID and EmployeeID. Display each OrderID, sales amount and salesperson name.

Q14. Display customer names and their OrderIDs, showing only customers who have matching orders.

Q15. Display each product name and its associated OrderIDs. Include the product category.

Q16. LEFT JOIN all Customers with Orders. Display all customers and their OrderIDs, including customers without orders.

Q17. Find customers who have no matching records in Orders.

Q18. Display all products that have no matching order records. Return ProductID, product name and category.

Q19. Use RIGHT JOIN to show all products and their matching OrderIDs, including products without orders.

Q20. Display orders with OrderStatus = Delivered, including the customer name and product name.

Q21. Join Orders, Customers and Products. Display OrderID, customer first name, product name, quantity and sales.

Q22. Display orders for customers from the USA. Show customer name, country, OrderID and sales.

Q23. Calculate total sales for each customer. Display CustomerID, customer first name and total sales.

Q24. Find customers whose total sales across their orders exceed 30. Display customer name and total sales.

Q25. Count the number of orders handled by each employee. Display EmployeeID, employee name and order count.

Q26. Use a SELF JOIN on Employees to display each employee's name alongside their manager's name. Keep employees without a manager.

Q27. For each product category, calculate total quantity sold and total sales.

Q28. Count orders for every customer, including customers with zero orders.

Q29. Compare Orders and OrdersArchive using OrderID. Show all IDs, including records that exist in only one table. Use FULL OUTER JOIN if supported.

Q30. Join Orders and OrdersArchive on OrderID. Display IDs where the current OrderDate differs from the archived OrderDate, returning both dates.

PART C

Mixed SQL Practice

Q31–Q40

Q31. Join Customers and Orders. Display each customer's name, country, total sales and total number of orders. Include only customers whose total sales exceed 20. Sort highest to lowest.

Q32. Join Products and Orders. For each product, calculate total quantity sold, total sales and average sales per order. Show only products whose total quantity sold is greater than 1.

Q33. Join Employees and Orders. Calculate total sales for each salesperson, considering only Delivered orders. Show salespeople whose delivered sales exceed 20.

Q34. Join Customers and Orders. Display each customer's name, score and total sales. Filter for customers whose score is at least 500 and total sales exceed 20.

Q35. From Orders, find orders placed between 2025-01-01 and 2025-02-28, inclusive. Display OrderID, OrderDate, status and sales. Sort by OrderDate.

Q36. Join Orders and OrdersArchive on OrderID. Find matching records where the Sales value differs. Display OrderID, current sales and archived sales.

Q37. Join Orders, Customers and Products. Find orders where Quantity is at least 2 and Sales is greater than 15. Display customer name, product name, quantity and sales.

Q38. From Employees, calculate the number of employees and average salary for each department. Show only departments with an average salary greater than 60,000. Sort by average salary descending.

Q39. Use a LEFT JOIN between Products and Orders. For each category, display the number of products and total sales generated by orders. Include categories even if they have no orders.

Q40. Find customers who have no Delivered orders. Include customers who have no orders at all. Display CustomerID and customer name.

PART D

Set Operators

Q41–Q50

Q41. Using UNION, create one unique list of CustomerID values appearing in Orders and OrdersArchive.

Q42. Using UNION ALL, combine all OrderID values from Orders and OrdersArchive and preserve duplicates.

Q43. Using INTERSECT, find ProductID values that appear in both Orders and OrdersArchive.

Q44. Using EXCEPT, find CustomerID values that appear in Orders but not in OrdersArchive.

Q45. Using EXCEPT, find OrderStatus values that appear in OrdersArchive but not in Orders.

Q46. Create one unique list of countries from Customers and departments from Employees using UNION. Rename the output column as Location_or_Department.

Q47. Combine FirstName values from Customers and Employees using UNION ALL and add a Source column showing Customer or Employee.

Q48. Find OrderID values that exist in both Orders and OrdersArchive, then display them in ascending order.

Q49. Using set operators, return OrderID values that exist in exactly one of the two order tables.

Q50. Create a combined report from Orders and OrdersArchive with Source_Label, OrderID, CustomerID and Sales, keeping all rows.

PART E

Advanced JOIN Practice

Q51–Q65

Q51. Use an INNER JOIN between Customers and Orders to show customers from Canada who have placed orders.

Q52. Use a LEFT JOIN to show every employee and the total sales generated by their orders, including employees with no orders.

Q53. Join Orders, Products and Customers. Show only orders for products in the Accessories category.

Q54. Use a SELF JOIN to find employees who share the same ManagerID. Display both employee names and avoid duplicate pairs.

Q55. Use a LEFT JOIN to find products that have never generated sales.

Q56. Join Employees with their managers using a SELF JOIN. Display employee name, employee department, manager name and manager department.

Q57. Join Orders and Customers, then calculate the average Sales for each country.

Q58. Join Orders and Products and return the top 3 products by total sales.

Q59. Use Customers LEFT JOIN Orders and find customers who placed more than 2 orders.

Q60. Join Orders, Customers and Employees. Display OrderID, customer name, salesperson name and Sales.

Q61. Use a FULL OUTER JOIN between Customers and Orders on CustomerID. Identify unmatched customers and unmatched order customer IDs.

Q62. Join Products with Orders and calculate total revenue per category, including categories with no orders.

Q63. Join Orders with OrdersArchive and show records where OrderStatus is the same but Sales is different.

Q64. Use a SELF JOIN on Employees to find each manager and count how many direct reports they have.

Q65. Join Customers, Orders, Products and Employees to create a complete sales transaction report with customer, product, salesperson, quantity and sales.

PART F

Subqueries

Q66–Q80

Q66. Find customers whose Score is greater than the average customer Score using a subquery.

Q67. Find employees whose Salary is greater than the average Salary of all employees.

Q68. Find products whose Price is greater than the average product Price.

Q69. Find orders whose Sales value is greater than the average Sales value of all orders.

Q70. Find customers who have placed at least one order using an IN subquery.

Q71. Find customers who have never placed an order using NOT IN or NOT EXISTS.

Q72. Find employees who have handled at least one Delivered order using EXISTS.

Q73. Find the employee or employees with the highest Salary using a subquery.

Q74. Find the product or products with the highest Price using a subquery.

Q75. Find customers whose total sales are greater than the average total sales per customer using a subquery.

Q76. Find products that have been ordered more than the average number of times per product.

Q77. Find orders whose Sales is greater than the average sales for their ProductID using a correlated subquery.

Q78. Find employees whose Salary is greater than the average salary in their own Department using a correlated subquery.

Q79. Find customers whose Score is the maximum within their Country using a correlated subquery.

Q80. Find the second-highest employee Salary using a subquery.

PART G

Aggregate Functions

Q81–Q90

Q81. Calculate total sales, average sales, minimum sales and maximum sales from Orders in one query.

Q82. Calculate total Quantity sold for each ProductID and sort from highest to lowest.

Q83. Find the number of distinct customers who have placed orders.

Q84. Calculate total sales for each OrderStatus and show only statuses with more than 2 orders.

Q85. Calculate average product Price for each Category and round it to 2 decimal places.

Q86. Find the country with the highest average customer Score.

Q87. Calculate total sales per month using OrderDate and the date functions supported by your SQL database.

Q88. Find the top 5 customers by total sales.

Q89. Calculate each employee department's minimum, maximum and average Salary.

Q90. Create a summary by product category showing number of products, average price and maximum price.

PART H

Advanced Mixed Challenges

Q91–Q100

Q91. Find customers whose total sales are above the overall average customer total sales, and display their name, country and total sales.

Q92.	Find the top salesperson in each department based on total Delivered sales. Use JOINs, aggregation and a subquery or window function if supported.
Q93.	Compare current Orders with OrdersArchive and classify each OrderID as Current Only, Archive Only, or In Both.
Q94.	Create a customer report showing CustomerID, name, country, number of orders, total sales and average order value. Include customers with zero orders.
Q95.	Find product categories where total sales are above the average total sales across all categories.
Q96.	Find managers whose team has generated total sales greater than 100. Use Employees, a SELF JOIN and Orders.
Q97.	Find customers who purchased products from more than one category.
Q98.	Find products that have never appeared in either Orders or OrdersArchive.
Q99.	Create a report of customers who have orders in both Orders and OrdersArchive.
Q100.	Create a final sales dashboard query showing country, category, number of orders, total quantity, total sales and average sales per order. Include only groups with total sales greater than 20.

KEEP PRACTISING. KEEP QUERYING. KEEP GROWING.

100 questions. Your next step is to solve them all.